AST 6215 Stars and the Galaxy

Problem Set #3

Due Oct 6, 2026 (4am)

Don't forget to list all of your collaborators on this assignment and resources used

1. White Dwarfs

A white dwarf may be approximated by a two zone model. Zone 1 is a helium interior composed of a non-relativistic, fully degenerate electron gas at constant temperature T_c with equation

$$P_e = K \rho^{5/3}$$

where K is a constant and P_e is the electron pressure. The very thin outer layers are composed of hydrogen, and obey an ideal gas equation of state with negligible radiation pressure. Energy transport in the outer layers occurs by radiative diffusion, with the opacity

$$\kappa = \kappa_0 \rho T^{-3.5}.$$

The transition between the interior and outer layers occurs when the total pressure given by the ideal gas law equals the electron pressure. Show that

a) That in the very thin outer layer the pressure is related to the temperature via

$$P^2 = \frac{64}{51} \frac{\pi a c G M}{\kappa_0 L} \left(\frac{R_{gas}}{\mu}\right) T^{8.5} = J T^{8.5}$$

where R_{gas} is the gas constant and M and L are the mass and luminosity of the white dwarf.
Hint: 'very thin' means that you can assume m=M, r=R and L_r=L throughout the outer layer.

b) That the temperature of the transition is

$$T_{tr} = \left(\frac{R_{gas}}{\mu}\right)^{10/7} K^{-6/7} J^{-2/7}$$

c) That the luminosity of the white dwarf is

$$L = \frac{64}{51} \frac{\pi a c G M}{\kappa_0} \left(\frac{R_{gas}}{\mu}\right)^{-4} K^3 T_c^{3.5}$$

d) The above expression for luminosity can be written as $L=C_1 M T^{3.5}$, where C_1 is a constant. The internal energy for a white dwarf is $U=C_2 M T_c$, with C_2 a constant. Show that, if the white dwarf has cooled from an initial temperature $T_0 >> T_c$ to its present temperature in time τ, then

$$\tau \propto \left(\frac{L}{M}\right)^{-5/7}$$

2. Main sequence

In class, we used the stellar structure equations and homology relations to show that for a main sequence star with an ideal gas equation of state that

$$T \sim \left(\frac{\mu}{R_{gas}}\right) \frac{G M}{R}$$

a) Assuming that the energy is being transported by radiation show that

$$L \propto \frac{a c}{\kappa} \left(\frac{\mu G}{R_{gas}}\right)^4 M^3$$

b) Assuming that the energy generation rate scales as $\varepsilon = \varepsilon \rho T^{\nu}$, show that

$$R^{3+\nu} \sim \frac{\epsilon_0 \kappa}{a c} \left(\frac{G \mu}{R_{gas}}\right)^{(\nu-4)} M^{(\nu-1)}$$

c) For stars dominated by the pp chain ν~4 and for stars dominated by the CNO cycle ν~20. Show that the density of stars dominated by the pp chain scales as $\rho \sim M^{-2/7}$, while the density of stars undergoing the CNO cycle scales as $M^{-34/23}$. Write at least two sentences explaining what this means for main sequence stars.

d) Using the fact that $L \sim 4 \pi R^2 \sigma T^4$, show that the slope of the main sequence should scale as

$$\log L = 5.6 \log T + \text{constant} \quad \text{(pp chain)}$$
$$\log L = 8.9 \log T + \text{constant} \quad \text{(CNO cycle)}$$

How does this compare to the observed slope of the main sequence (you can use your YREC models to estimate this)?

e) You can rearrange the previous equations to get that the $T_{central} \sim M^{4/(\nu+3)}$. If the central temperature drops below 4×10^6 K, the star will not be able to fuse hydrogen. Estimate what mass this happens at. How does this compare to the boundary between stars and brown dwarfs?

f) The maximum Luminosity stars can have should be something like the Eddington Luminosity

$$L_{Edd} = \frac{4 \pi c G M}{\kappa}$$

Show that this implies a maximum mass for stars. Substituting in the numbers, you would get something like a maximum mass of 114 Msun. Do you think that this should be a firm limit?